

Collections – List and Sets

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COURSE CODE: CSA0904

COURSE NAME: PROGRAMMING FOR JAVA

1

```
import java.util.*;
```

```
class Student {
```

```
    public static void main(String args[]) {
```

```
        ArrayList<String> wait = new ArrayList<>();
```

```
        ArrayList<String> conf = new ArrayList<>();
```

```
        wait.add("Rajan");
```

```
        wait.add("ramesh");
```

```
        wait.add("suresh");
```

```
        wait.add("rajesh");
```

```
        conf.add("suresh");
```

```
        conf.add("somesesh");
```

```
        conf.add("rajesh");
```

```
        Scanner sc = new Scanner(System.in);
```

```
        String dropout = sc.nextLine();
```

```
        if (wait.contains(dropout)) {
```

```

        System.out.println("found");
    } else {
        System.out.println("Not Found");
    }

    Collections.sort(conf);

    System.out.println(conf);

    conf.addAll(wait);

    System.out.println(conf);

    sc.close();
}
}

```

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and the user's name (SELVA PRABHU A K) with a dropdown menu. The main content area is divided into three sections: a sidebar on the left with a course enrollment system status, a central problem description, and a coding area on the right.

Course Enrolment system: 42 Course Enrolment system, 10 marks, 2/2 answered.

Problem Description:

1. create 2 list waiting list and confirmed list
2. get a dropout student name from keyboard - search is it in the waiting list (list contains "Ramesh")
3. if so remove from the list
4. sort the confirmed students list
5. merge waiting list with confirmed list

Your Code:

```

1 import java.util.*;
2 class Student{
3     public static void main(String args[]){
4         ArrayList<String>wait = new ArrayList<>();
5         ArrayList<String>conf=new ArrayList<>();
6         wait.add("Rajan");
7         wait.add("ramesh");
8         wait.add("suresh");
9         wait.add("rajesh");
10        conf.add("suresh");
11        conf.add("somesesh");
12        conf.add("rajesh");
13        Scanner sc = new Scanner(System.in);
14        String dropout = sc.nextLine();
15        if(wait.contains(dropout)){
16            System.out.println("found");
17        }
18        else{
19            System.out.println("Not Found");

```

Input: ramesh

Output:

```

found
[rajesh, somesh, suresh]
[rajesh, somesh, suresh, Rajan,
ramesh, suresh, rajesh]

```

2

```
import java.util.*;
```

```

class Course {

    public static void main(String args[]) {

```

```
HashSet<String> java = new HashSet<>();
```

```
HashSet<String> python = new HashSet<>();
```

```
java.add("Rama");
```

```
java.add("Arun");
```

```
java.add("Kumar");
```

```
java.add("Priya");
```

```
java.add("Anu");
```

```
python.add("Rama");
```

```
python.add("Hari");
```

```
python.add("Vijay");
```

```
python.add("Priya");
```

```
python.add("Kavi");
```

```
if (java.contains("Rama"))
```

```
    System.out.println("Rama Found in Java");
```

```
if (python.contains("Rama"))
```

```
    System.out.println("Rama Found in Python");
```

```
HashSet<String> union = new HashSet<>(java);
```

```
union.addAll(python);
```

```
System.out.println("Union = " + union);
```

```
HashSet<String> inter = new HashSet<>(java);
```

```
inter.retainAll(python);
```

```
System.out.println("Intersection = " + inter);
```

```
TreeSet<String> tree = new TreeSet<>(union);
```

```
System.out.println("TreeSet = " + tree);
```

```
LinkedHashSet<String> link = new LinkedHashSet<>(union);
```

```
System.out.println("LinkedHashSet = " + link);
```

```
}
```

```
}
```

The screenshot displays the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", the subject "JAVA PROGRAMMING", and the user profile "SELVA PRABHU A K" with ID "1924212248A".

On the left, a "QUESTIONS" sidebar lists two questions: "Q1 Student management system" (10 marks) and "Q2 Course Enrolment system" (10 marks). The "Q2" question is selected, and it shows "2/2 answered".

The main area for "Q2 Course Enrolment system" (10 marks) contains the following text:

You are developing a course enrollment system where each course can have multiple students, but each student can enroll in a course only once. The system needs to prevent duplicate enrollments and perform various set operations. perform the following

1. create 2 courses javacourse and python course
2. add 5 students in each
3. check student "rama" is there, if so remove from the list
4. find the union, intersection of both list
5. create treeset, linkedhashset to maintain order and alphabetical order

Below the text, there are three sections:

- Your Code:** A text area containing the following Java code:

```
1 import java.util.*;
2
3 class Course {
4     public static void main(String args[]) {
5
6         HashSet<String> java = new HashSet<>();
7         HashSet<String> python = new HashSet<>();
8
9         java.add("Rama");
10        java.add("Arun");
11        java.add("Kumar");
12        java.add("Priya");
13        java.add("Anu");
```
- Input:** A text input field labeled "stdin input".
- Output:** A text area showing the program's output:

```
Rama Found in Java
Rama Found in Python
Union = [Hari, Vijay, Kumar,
Priya, Rama, Arun, Anu, Kavi]
Intersection = [Priya, Rama]
HashSet = [Arun, Arun, Hari, Kavi]
```

